Deep Learning and Reinforcement Project

Dataset: Ventilator Pressure Prediction

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1. Main Objective

The dataset contains information regarding ventilator output and input using artificial bellows test lung via a respiratory circuit. We can see the diagram process on Fig.1. The diagram shows a state variable of airway pressure detected by the pressure sensor and the 2 control valves and, which entails:

- The inspiratory valve: as control input with a continuous variable from 0 to 100 representing the percentage the inspiratory solenoid valve is open to let air into the lung (i.e., 0 is completely closed and no air is let in and 100 is completely open).
- The expiratory valve: as second control input is a binary variable representing whether the exploratory valve is open (1) or closed (0) to let air out.

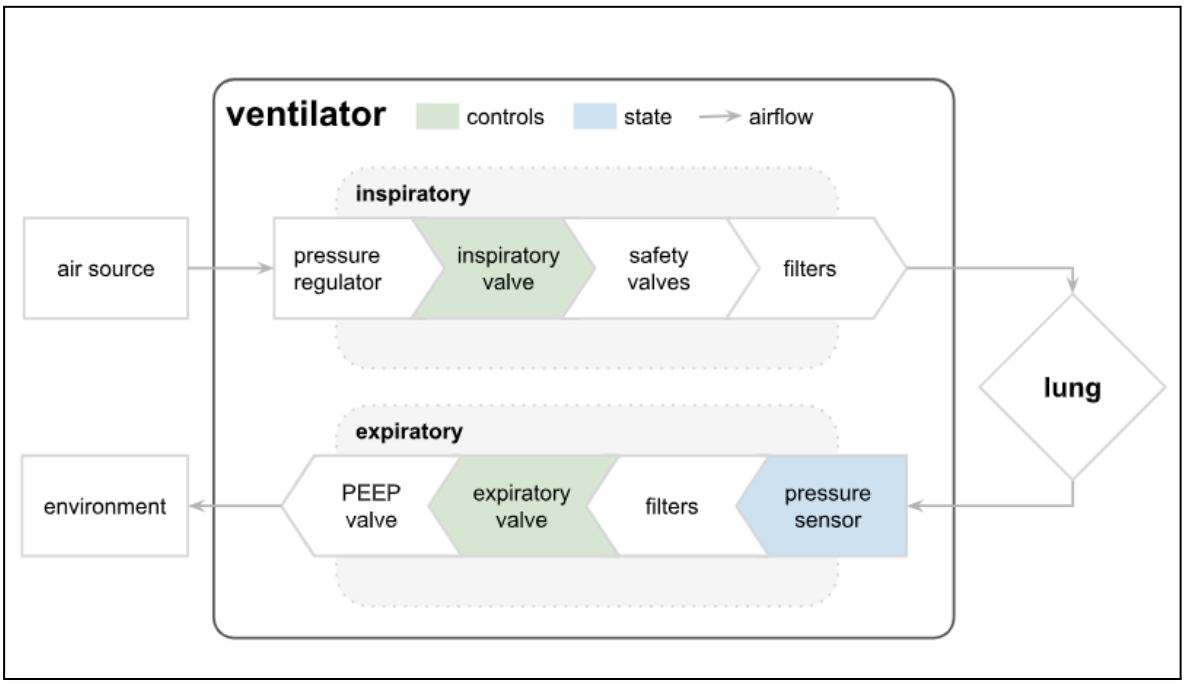

Fig.1. Flow Diagram of a ventilator

The dataset comprises a total of **6,036,000 time series with 7 features and 1 target column.** Each time series represents an approximately 3-second breath. The creator of the dataset intended to simulate a ventilator connected to a sedated patient's lung by predicting the ventilator pressure prediction.

With the creator objective and the information of the dataset, we can predict the pressure using below methods of deep learning then use **MAE** to decide the most optimal model. Below is the deep learning method we'll use:

- Multi-Layer Perceptron and Bidirectional Long Short-Term Memory (MLP-BiLSTM)
- Bidirectional Long Short-Term Memory + Bidirectional Gated Recurrent Unit (BiLSTM+BiGRU)
- Multi-Scale Convolutional Neural Network + Bidirectional Long Short-Term Memory (Multi-Scale CNN+BiLSTM)

2. Exploration Data, Data Cleaning, and Feature Engineer

2.1 Exploration Data

2.1.1 Gathering Information

We need to get a snippet of the dataset first through data.sample(). to get a snippet of the dataset and *data.info()* in Fig. 2. to understand the datatype, total entry, and NaN existence.

```
df_train.sample(5)
```

	id	breath_id	R	C	time_step	u_in	u_out	pressure
3369583	3369584	70242	50	10	2.104927	1.581599	1	4.782960
3595191	3595192	74946	5	20	2.266899	4.903171	1	6.540513
5221648	5221649	108719	50	50	1.653128	2.848099	1	6.189002
1965130	1965131	40777	20	10	0.338608	9.985675	0	17.859159
553530	553531	11462	5	20	0.318351	44.847521	0	18.562180

Fig.1. data.sample()

```
df_train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6036000 entries, 0 to 6035999
Data columns (total 8 columns):
 #   Column     Dtype
---  ------     -----
 0   id         int64
 1   breath_id  int64
 2   R          int64
 3   C          int64
 4   time_step  float64
 5   u_in       float64
 6   u_out      int64
 7   pressure   float64
dtypes: float64(3), int64(5)
memory usage: 368.4 MB
```

Fig.2. data.info()

The dataset's creator also has shared explanation for each variables, which entails:

- **id**: globally-unique time step identifier across an entire file
- **breath_id**: globally-unique time step for breaths
- **R**: lung attribute indicating how restricted the airway is (in cmH2O/L/S).
- **C**: lung attribute indicating how compliant the lung is (in mL/cmH2O).
- **time_step**: the actual time stamp.
- **U_in**: the control input for the inspiratory solenoid valve. Ranges from 0 to 100.
- **U_out**: the control input for the exploratory solenoid valve. Either 0 or 1.
- **pressure**: the airway pressure measured in the respiratory circuit, measured in cmH2O.

2.1.2 Insight

Based on Fig. 1. And Fig. 2., the dataset has:

1. From Fig. 2. data.info(), there are a total 8 columns, which entails 7 columns of features.
2. Based on the dataset's snippet Fig.1. All the values are in number and all the data types are mostly showing integer or float.

2.2 Data Cleaning

In this dataset we found no missing value and duplicated value as shown in Fig.3.

```
[TRAIN] Missing values: NONE ✓
[TRAIN] Exact duplicate rows     : 0
[TRAIN] Duplicate breath+timestep: 0
[TRAIN] Outliers (pressure): NONE ✓
```

Fig.3. Missing Value, Duplicated Value, and Outlier Handling

3. Data Visualization

First we need to check the statistical condition in the ‘Pressure’ column in Fig.4. We found that there are anomalies where a huge gap exists between the max value and the Q3(75%) value.

```
Basic stats (pressure - target):
count    6036000.000
mean          11.220
std            8.110
min           -1.896
25%            6.330
50%            7.033
75%           13.641
max           64.821
Name: pressure, dtype: float64
```

Fig.4. Missing Value, Duplicated Value, and Outlier Handling

In Fig.5. show the pressure distribution graph. Two distinct pressure regimes: inspiratory (scored) is higher and more variable than expiratory. This is why the loss function uses u_out as a mask. They are fundamentally different signal types.

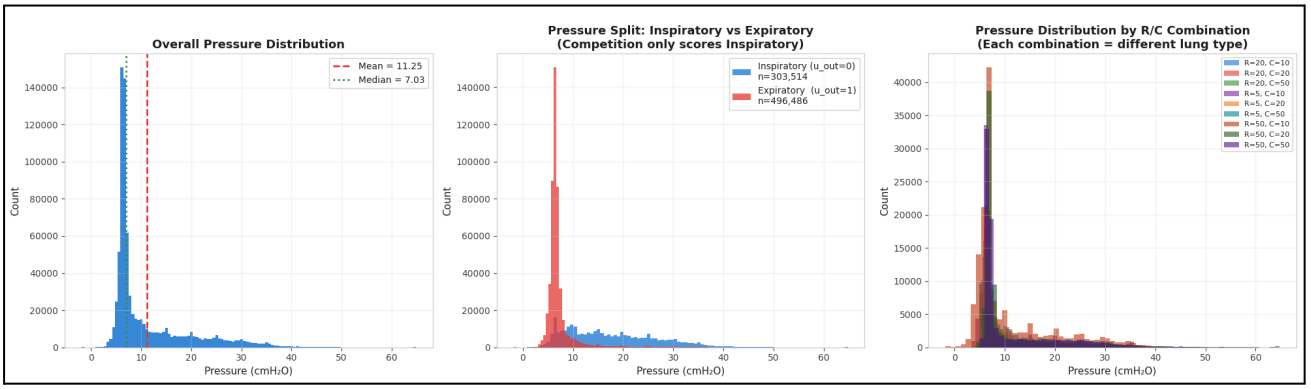

Fig.5. Pressure Distribution

In Fig.6. show a Pressure DistributionInter-Feature Correlation Heatmap. We can find features u_in, u_in_lag1, u_in_lag2 are highly correlated with each other and feature cumsum_u_in is correlated with u_in but not identical and it captures accumulated history, not instantaneous value. However, features R and C are uncorrelated by design (experimental controls).

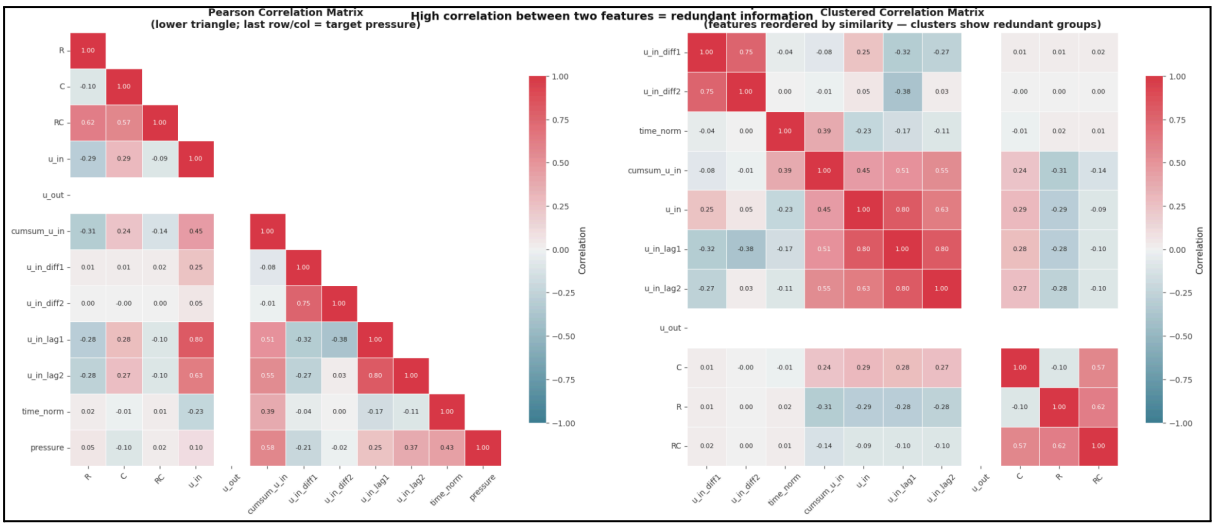

Fig.6. Pressure Distribution

Based on those graphs, we can take several insight and keypoint before running the model:

1. cumsum_u_in is the most important single feature.
2. Temporal lag structure is strong by LSTM/GRU hidden states and TCN dilations are well-justified.
3. R/C have discrete, step-like effects where the MLP pre-processor is critical to encode them non-linearly before the LSTM.
4. Non-linear relationships exist for diff1/diff2/lags.
5. The dataset is well-structured (no obvious outliers or missing value issues) and the training should be stable.

4. Implementing and Evaluation Model

Before running all the models, we need to create a blueprint for each model to define its architecture, then training each model with it. We'll run each model in 50 epochs.

- **Multi-Layer Perceptron and Bidirectional Long Short-Term Memory (MLP-BiLSTM)**

  The MLP translates raw sensor numbers into a richer language, then the BiLSTM reads the breath forwards and backwards like rereading a sentence, and outputs one predicted pressure number for each of the 80 timesteps. This model will act as a baseline and learn 463,617 parameters. Find Fig.7. for its evaluation matrix

```
============================================================
TRAINING MODEL 1: MLP + BiLSTM (Baseline)
============================================================

── Fold 1/5 ──────────────────────────
  Train: 60,360 | Val: 15,090 | Params: 463,617
  Epoch   1/50 | train=12.5314 | val=5.3788 | mae=5.4069 | lr=4.01e-04 | patience=0/7
  Epoch  10/50 | train=1.5500 | val=1.4572 | mae=1.4643 | lr=8.42e-04 | patience=1/7
  Epoch  20/50 | train=1.2202 | val=1.2243 | mae=1.2304 | lr=6.32e-04 | patience=1/7
  Epoch  30/50 | train=1.0523 | val=1.0319 | mae=1.0367 | lr=4.21e-04 | patience=2/7
  Epoch  40/50 | train=0.9250 | val=0.8604 | mae=0.8641 | lr=2.11e-04 | patience=0/7
  Epoch  50/50 | train=0.8523 | val=0.8318 | mae=0.8353 | lr=0.00e+00 | patience=0/7
  Best Val MAE (fold 1): 0.8353

>>> CV MAE (1 folds): 0.8353 ± 0.0000
```

Fig.7. Evaluation Matrix

- **Bidirectional Long Short-Term Memory + Bidirectional Gated Recurrent Unit (BiLSTM+BiGRU)**

  In this section, we combine 2 models where the BiLSTM reads the breath carefully and takes detailed notes, then the BiGRU skims those notes to extract the key patterns, and together they output one predicted pressure per timestep. This model will learn from 825,857 parameters. Find Fig.8. for its evaluation matrix.

```
============================================================
TRAINING MODEL 2: BiLSTM + BiGRU Stack
============================================================
  Train breaths: 60,360 (60,360 rows)
  Val   breaths: 15,090 (15,090 rows)
  Parameters   : 825,857
  Epoch   1/50 | train=10.7441 | val=3.9690 | mae=3.9936 | lr=4.01e-04 | patience=0/7
  Epoch  10/50 | train=1.7939 | val=1.5125 | mae=1.5222 | lr=8.42e-04 | patience=0/7
  Epoch  20/50 | train=1.3186 | val=1.1043 | mae=1.1088 | lr=6.32e-04 | patience=3/7
  Epoch  30/50 | train=1.1599 | val=0.9341 | mae=0.9394 | lr=4.21e-04 | patience=3/7
  Epoch  40/50 | train=1.0604 | val=0.7840 | mae=0.7879 | lr=2.11e-04 | patience=0/7
  Epoch  50/50 | train=1.0006 | val=0.7251 | mae=0.7285 | lr=0.00e+00 | patience=0/7
  Best Val MAE : 0.7285
```

Fig.8 .Evaluation Matrix

- **Multi-Scale Convolutional Neural Network + Bidirectional Long Short-Term Memory (Multi-Scale CNN+BiLSTM)**

  This model takes three measures to look at the breath simultaneously at different levels (close-up, medium, wide), their combined picture is handed to the BiLSTM to understand the sequence over time, and the output is one predicted pressure per timestep. This model will learn from 464,321 parameters. Find Fig.9. for its evaluation matrix.

```
============================================================
TRAINING MODEL 3: Multi-Scale CNN + BiLSTM
============================================================
  Train breaths: 60,360 (60,360 rows)
  Val   breaths: 15,090 (15,090 rows)
  Parameters   : 464,321
  Epoch   1/50 | train=12.5185 | val=4.5327 | mae=4.5570 | lr=4.01e-04 | patience=0/7
  Epoch  10/50 | train=1.4343 | val=1.1225 | mae=1.1296 | lr=8.42e-04 | patience=0/7
  Epoch  20/50 | train=1.1883 | val=0.9770 | mae=0.9819 | lr=6.32e-04 | patience=3/7
  Epoch  30/50 | train=1.0761 | val=0.8083 | mae=0.8127 | lr=4.21e-04 | patience=0/7
  Epoch  40/50 | train=1.0168 | val=0.7647 | mae=0.7674 | lr=2.11e-04 | patience=0/7
  Epoch  50/50 | train=0.9641 | val=0.7060 | mae=0.7089 | lr=0.00e+00 | patience=2/7
  Best Val MAE : 0.7035
```

Fig.9.Evaluation Matrix

5. Model Recommendation

Based on the training results, **Multi-Scale CNN + BiLSTM is the recommended model** with the best Val MAE of 0.7035, outperforming BiLSTM + BiGRU (0.7285) and MLP + BiLSTM (0.8353). The CNN component excels at extracting multi-scale local patterns from the ventilator waveform, short kernels catch sharp u_in spikes while long kernels capture the slower pressure build-up curve, which is exactly the kind of multi-resolution signal present in breath sequences. BiLSTM + BiGRU ranks second at 0.7285, benefiting from its dual recurrent architecture that captures both forward and backward temporal dependencies simultaneously. MLP + BiLSTM ranks last at 0.8353, which is expected since the MLP front-end treats each timestep somewhat independently before passing to BiLSTM, losing some fine-grained sequential structure early in the pipeline. For production deployment on Kaggle's ventilator pressure prediction, Multi-Scale CNN + BiLSTM offers the best accuracy-to-complexity tradeoff with only 464,321 parameters and nearly identical to MLP + BiLSTM (463,617) but with significantly better MAE.

6. Insight and Key Findings: Dataset Objective

All three models show a consistent and healthy learning pattern on training loss and validation loss both decrease smoothly across 50 epochs without signs of divergence, confirming that the 80/20 breath-id-aware split is working correctly and no breath leakage exists between train and val sets. The gap between train MAE and val MAE narrows significantly by epoch 50 across all models, suggesting the models generalise well rather than memorising specific breath waveforms. A key finding is that all models struggle most in the early epochs (epoch 1 MAE > 4.0 for all), which reflects how complex the relationship is between u_in, R, and C parameters and the resulting airway pressure in the models need many gradient steps before learning the underlying respiratory mechanics. Multi-Scale CNN + BiLSTM converges to the lowest value MAE of 0.7035, demonstrating that **local waveform shape features** (captured by CNN kernels) are highly predictive of pressure, on top of the temporal sequence modelling provided by BiLSTM. The ventilator dataset's core challenge predicting continuous pressure values across 80 timesteps per breath under varying lung resistance (R) and compliance (C) conditions is best handled by architectures that can simultaneously model both local signal morphology and long-range temporal dependencies.

7. Model Plan of Improvement

The most impactful next step is to implement an attention mechanism on top of the BiLSTM layer in all three models, allowing the model to dynamically focus on the most pressure-critical timesteps (particularly the inspiratory phase transition) rather than weighting all timesteps equally. Feature engineering can be further enriched by adding RC interaction terms, breath-level statistics (per-breath mean u_in, peak u_in, u_in integral), and positional encoding of timestep within breath and features that directly encode the physics of the lung compliance equation. For Multi-Scale CNN + BiLSTM specifically, increasing the number of CNN branches from 3 to 5 with kernel sizes spanning 3 to 21 would capture even broader multi-scale patterns, and adding residual skip connections between CNN output and BiLSTM output

would help preserve raw signal information throughout the deep pipeline. Training improvements include switching from a linear warmup-decay scheduler to a cosine annealing with warm restarts scheduler, which has been shown to help sequence models escape local minima more effectively on time-series regression tasks. Finally, assembling all three models via a simple weighted average of their predictions (weighted by inverse val MAE) would likely push the final score below 0.70 MAE without requiring any additional training, as each model captures complementary aspects of the pressure waveform.